

# W-Fibers as Constraint-Projection Relations: A Non-Causal Fiber-Like Formalism for Global Admissibility

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## Abstract

This paper formalizes the concept of *W-fibers* as non-causal, relational structures arising from global admissibility constraints rather than physical connections or interactions. Building on

the Dimension-W framework, W-fibers are defined as local expressions of compatibility relations between a global constraint space and realized systems such as spacetime histories, computational executions, or statistical models. Using bundle-like notation without ontological commitment to fiber objects, we show that W-fibers function as projection relations that preserve consistency without mediating influence. This approach resolves prior ambiguities associated with fiber metaphors, avoids metaphysical leakage, and unifies constraint-based reasoning across physics, computation, and probability theory.

# 1. Introduction

Many theoretical frameworks implicitly rely

on global constraints that restrict which local states or histories may exist. These constraints often manifest uniformly across a system, giving rise to the impression of laws, regularities, or persistent structures. In prior work, these constraints were described using the language of *W-fibers*, suggesting a fiber-like relationship between a global admissibility space (Dimension- $W$ ) and local realizations.

However, without careful formalization, fiber language risks being misinterpreted as implying physical connections, causal influence, or hidden interactions. The purpose of this paper is to provide a precise, non-causal definition of  $W$ -fibers grounded in standard relational and constraint-based formalisms. We demonstrate that  $W$ -fibers are best understood as compatibility relations instantiated locally, not as entities that act

upon or connect systems.

## 2. Background: Dimension-W as Admissibility Space

Dimension-W is defined as the space of globally admissible structures for a given system. It does not represent a physical dimension, causal domain, or dynamical layer. Instead, it encodes the total set of constraints that determine which configurations, histories, or executions are logically or mathematically permitted. Examples of Dimension-W analogues include:

- The space of admissible trajectories satisfying an action principle in classical mechanics (Goldstein, Poole, & Safko, 2002; Landau & Lifshitz, 1976).

- The set of all valid computational executions consistent with a formal semantics (Winskel, 1993; Sipser, 2012).
- The space of probability measures satisfying normalization and coherence axioms (Kolmogorov, 1956; Jaynes, 2003).

In each case, the admissibility space exists as a logical structure, not as a physical object.

### 3. Definition of Relation in the W-Fiber Context

To avoid ambiguity, we adopt the following working definition:

*A relation* is a formal compatibility

condition between elements of two descriptive spaces, specifying which combinations are admissible without implying interaction, influence, or exchange.

Formally, let:

- $W$  denote the global admissibility or constraint space.
- $S$  denote the space of local states, histories, or realizations.

A relation  $R$  is a subset of the Cartesian product  $W \times S$  such that:

$(w, s) \in R$  if and only if  $s$  is admissible under  $w$ .

This relation is descriptive rather than dynamical. It does not encode causation, information flow, or mutual action.

## 4. W-Fibers as Projection Relations

Within this framework, a W-fiber is defined as the local instantiation of the relation  $R$  at a given point or region in  $S$ . Informally, it answers the question: *Which global constraints are consistent with this local state?*

Using bundle-like notation purely as a representational aid:

- The base space corresponds to  $S$  (local realizations).
- The total space corresponds to  $W \times S$  restricted by  $R$ .
- The fiber over a local state  $s \in S$  is the set  $\{ w \in W \mid (w, s) \in R \}$ .

No ontological claim is made that fibers exist as objects or structures in reality. The notation simply organizes admissibility information.

## **5. Non-Causality and the Absence of Interaction**

A central feature of W-fibers is their explicitly non-causal character. Global constraints do not act on local states, nor do local states influence global constraints. Instead, inconsistent states never arise.

This mirrors familiar cases:

- A trajectory that violates boundary conditions is not dynamically corrected; it is simply not a solution.

- A program that violates type constraints does not run incorrectly; it fails to compile.
- A probability distribution that fails normalization is not improbable; it is invalid.

W-fibers encode the same logic of exclusion rather than enforcement.

## 6. Relation to Physical Laws and Uniformity

The apparent uniformity of physical laws can be understood as the persistence of admissibility relations across all local realizations. W-fibers express how the same global constraints are reflected locally without invoking fields, forces, or hidden variables.

Under this interpretation, laws are invariants of admissibility rather than commands imposed on spacetime. This aligns with action-based formulations in physics (Arnold, 1989; Feynman & Hibbs, 1965) and with modern discussions of top-down constraint and emergence (Ellis, 2014).

## **7. Discussion**

Recasting W-fibers as constraint-projection relations clarifies their role and removes the need for metaphysical interpretations. Fiber language is retained only insofar as it aids visualization and formal organization. All causal, temporal, and spatial dynamics remain internal to the realized system.

This framework is compatible with a wide range of models and does not privilege cosmological, cognitive, or physical interpretations. It provides a general tool for understanding how global consistency gives rise to local structure.

## 8. Conclusion

W-fibers are not connections, channels, or mechanisms. They are relational expressions of admissibility that appear locally wherever a globally constrained system is realized. By formalizing W-fibers in this way, we preserve the explanatory power of fiber metaphors while eliminating ontological and causal ambiguities. This positions the W-fiber framework as a unifying, conservative extension of standard constraint-based reasoning.

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